Bode Part 3: Wrapping Up the Six Building Blocks

EGR 345

GVSU

Outline

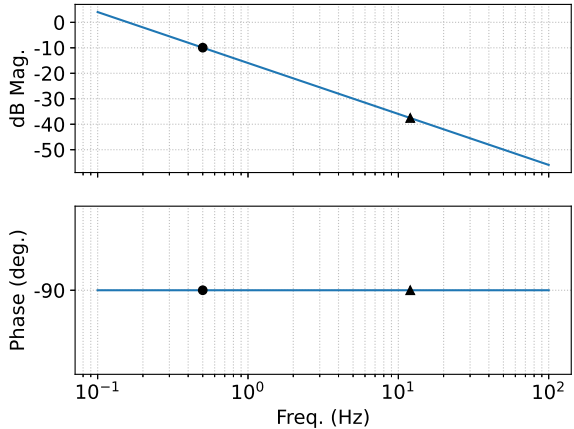
- review
- last two building blocks
- combining building blocks
- examples

Review (Slide 2)

What does this Bode plot look like?

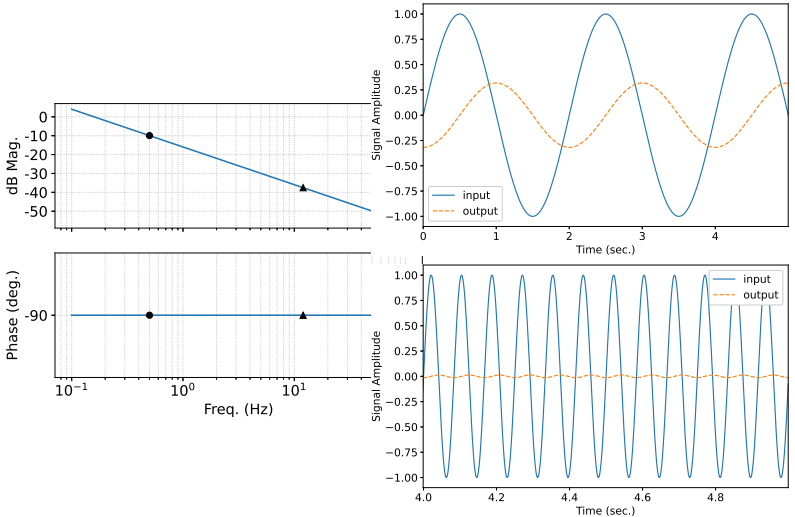
$$G_1(s) = \frac{1}{s}$$

G_1 Bode



- what does the steady-state fixed sine response look like at the two marked frequencies?

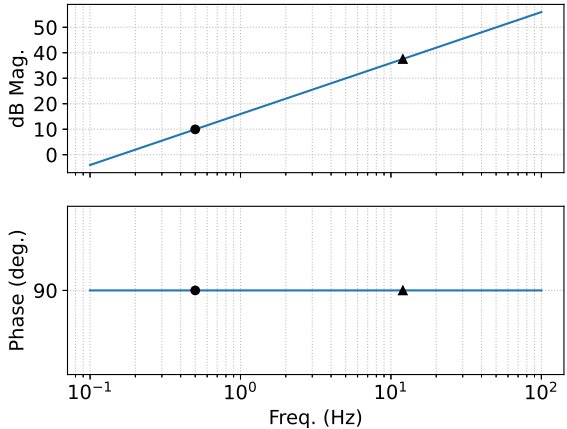
G_1 Bode and steady-state fixed sine



What does this Bode plot look like?

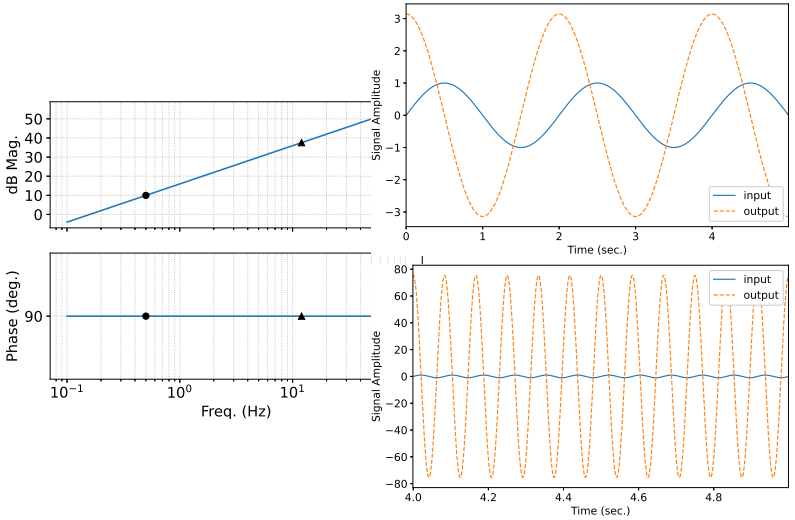
$$G_2(s) = s$$

G_2 **Bode**



- what does the steady-state fixed sine response look like at the two marked frequencies?

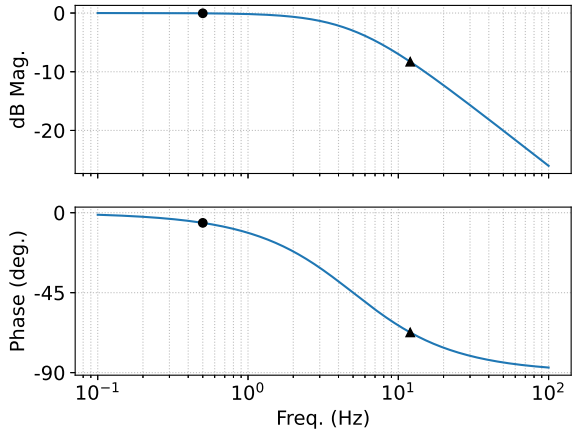
G_2 Bode and steady-state fixed sine



What does this Bode plot look like?

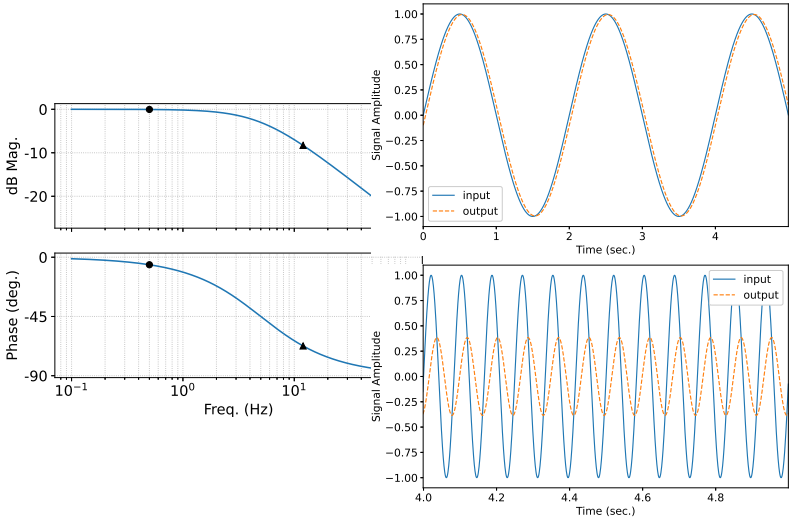
$$G_3(s) = \frac{p}{s+p}$$

G_3 Bode



- what does the steady-state fixed sine response look like at the two marked frequencies?

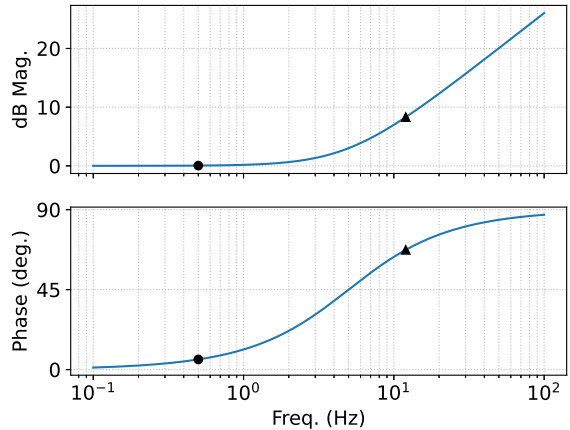
G_3 Bode and steady-state fixed sine



What does this Bode plot look like?

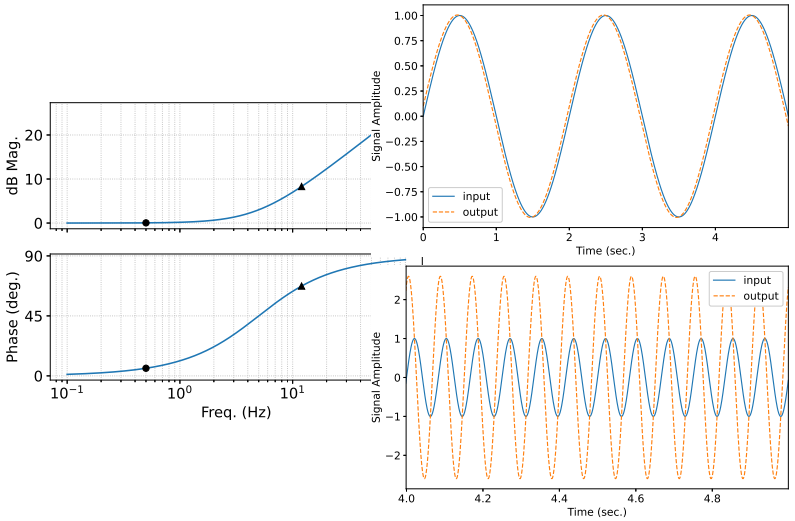
$$G_4(s) = \frac{s+a}{s}$$

G_4 Bode



- what does the steady-state fixed sine response look like at the two marked frequencies?

G_4 Bode and steady-state fixed sine



Building Blocks 5 and 6

Building block 5: second-order (complex) poles

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- What happens to the magnitude and phase of $G_5(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

Building Block 6: Under-Damped Second-Order Zero

$$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

- What happens to the magnitude and phase of $G_6(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

Bode Sketching Algorithm

- find all poles and zeros
  - convert them to Hz
  - sort them from lowest to highest frequency
- identify the low frequency behavior
  - pure int, pure diff. or neither
    - first or second order int. or diff. possible
- work your way up in frequency
- when you reach a pole or zero, increment or decrement the slope **and** phase
- continue until you are above the highest frequency pole or zero
- double check the high frequency slope and phase
  - slope: $-20(n-m)$ dB/dec
  - phase: $-90^\circ(n-m)$

Bode Sketching Example 1

$$G_{ex1} = \frac{p}{s(s+p)}$$

assume $p = 10 \cdot 2 \cdot \pi$

Bode Sketching Example 2

$$G_{ex2} = \frac{p_1 \cdot p_2}{(s+p_1)(s+p_2)}$$

assume $p_1 = 0.1 \cdot 2 \cdot \pi$ and $p_2 = 25 \cdot 2 \cdot \pi$

Random Bode Generator

- arguably the most instructive Python code I have ever written